

USE-CASE FLOW SPECIFICATION

Digital Art Commission & Watermarking Portal

Problem Statement #58 | PES University – CSE | SRN: PES1UG24CS916

1. Use Case Overview

Use Case ID: UC-04

Use Case Name: Process Milestone Payment & Release High-Res File

Primary Actor: Client Buyer

Secondary Actors: Digital Artist, Payment Gateway

Trigger: Client Buyer clicks **[Pay & Unlock]** on a completed milestone.

Related FR: FR-004

Priority: High

2. Preconditions

PC-1: The Client Buyer is authenticated and logged into the portal.

PC-2: The Digital Artist has uploaded the watermarked WIP draft for the milestone and marked it as ready for review.

PC-3: The Client Buyer has reviewed and approved the WIP draft.

PC-4: A valid payment method (credit/debit card or UPI) is registered with the Client Buyer's account.

3. Postconditions

PSC-1: Payment is recorded and a receipt is emailed to the Client Buyer.

PSC-2: The system generates an expiring signed S3 URL (valid 60 minutes) for the high-res source file.

PSC-3: The Digital Artist is notified and the milestone is marked **"Paid"** in the commission dashboard.

PSC-4: If the milestone is the final one, the commission status changes to **"Completed."**

4. Main Success Scenario

1. Client Buyer navigates to the commission dashboard and selects the approved milestone.
 2. System displays milestone details (WIP preview thumbnail, agreed price, remaining free revisions).
 3. Client Buyer clicks the **[Pay & Unlock]** button.
 4. System presents the payment summary screen: milestone title, amount, registered payment method.
 5. Client Buyer confirms payment by clicking **[Confirm Payment]**.
 6. System sends the payment request to the integrated Payment Gateway.
 7. Payment Gateway authorises the transaction and returns a success event with a transaction ID.
 8. System marks the milestone as **"Paid,"** records the transaction ID, and sends the Client Buyer a payment receipt via email.
 9. System generates an expiring signed S3 URL for the high-resolution source file (TTL: 60 minutes).
 10. System displays the **[Download High-Res File]** button with the signed URL embedded.
 11. Client Buyer clicks the button and the file downloads successfully.
 12. System notifies the Digital Artist via email/in-app notification that the milestone is paid.
 13. Use case ends successfully.
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5. Alternate Flows

AF-1: Payment Declined by Gateway

7a. Payment Gateway returns a decline event (insufficient funds, card expired, etc.).

7a1. System displays a **"Payment Failed"** banner with the decline reason.

7a2. System prompts the Client Buyer to retry with the same or a different payment method.

7a3. If the Client Buyer retries — return to Step 5.

7a4. If payment fails twice consecutively, the system locks the payment attempt for 15 minutes and sends a notification email to the Client Buyer. Milestone status remains **"Pending Payment."** Use case ends in failure.

AF-2: Client Buyer Does Not Confirm (Session Timeout)

5b. Client Buyer is inactive for 10 minutes on the payment summary screen.

5b1. System displays a timeout warning: session expiring in 60 s.

5b2. If no action is taken, the session expires, the payment summary is dismissed, and the Client Buyer is redirected to the commission dashboard. No charge is made. Use case ends.

AF-3: Signed URL Expires Before Download

11c. Client Buyer attempts to download after the 60-minute TTL has elapsed.

11c1. System returns an **"Access Denied - Link Expired"** message.

11c2. Client Buyer clicks **[Generate New Link]**.

11c3. System validates that payment is confirmed, then generates a new 60-minute signed S3 URL.

11c4. Client Buyer downloads the file. Use case continues from Step 11.